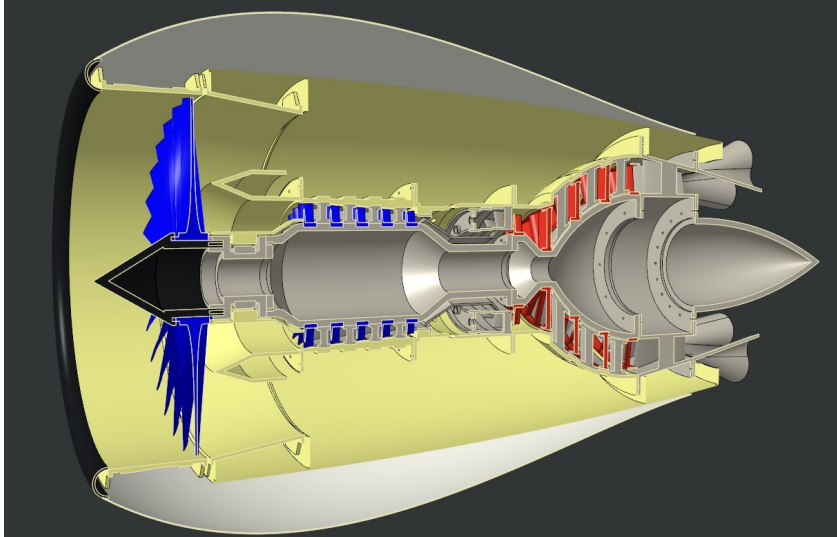


Aaryan Lath

aaryanlath.com | linkedin.com/in/aaryan-lath | (312) 838-5669 | aaryanlath05@gmail.com | 7604 Memphis Drive, Plano, TX

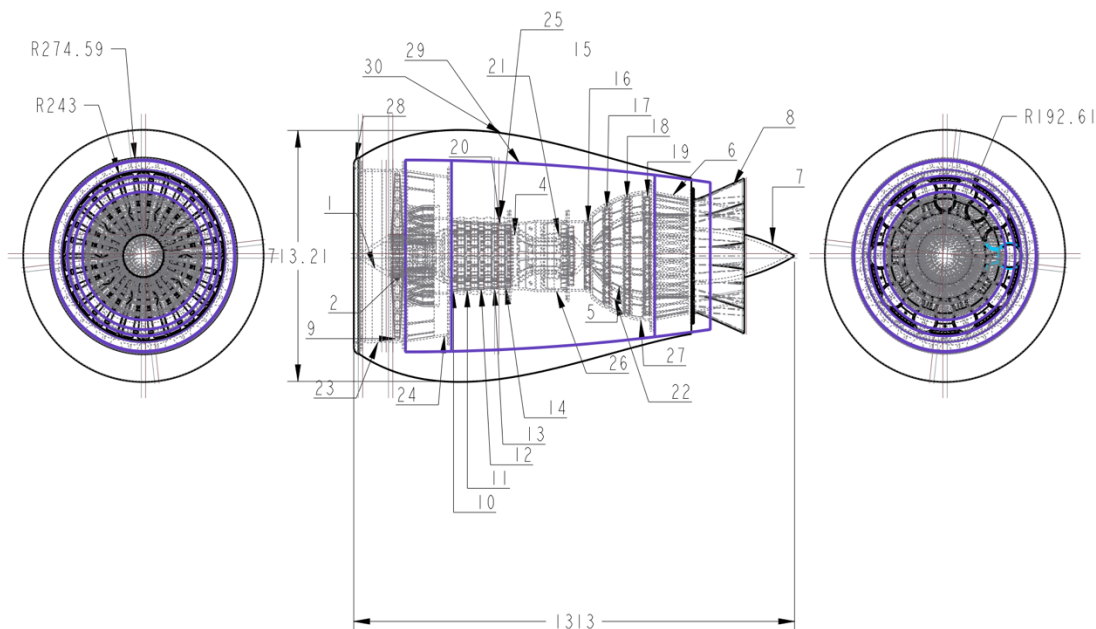
Jet-Engine



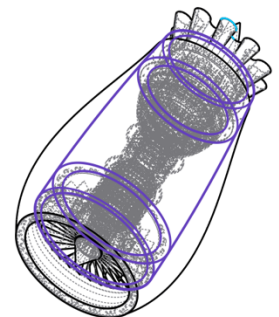
Top-Down assembly design of a turbofan engine for understanding individual components.

How:

- Employed **CREO Parametric** for the model
- Applied **GD&T** on all components
- Completed a complete model with **80 hours** of modelling.



ITEM	PC. NO.	QTY.
1	NOSE-CONE	1
2	PRIMARY-TURBINE-BUSHING	1
3	BEARING-ONE	1
4	CENTRAL-MOUNTING-SHAFT	1
5	REAR-MOUNTING-SHAFT	1
6	REAR-CASE	1
7	END-CONE	1
8	EXHAUST-CASE	1
9	PRIMARY-TURBINE-BLADES	1
10	AIR-TURBINE-ONE	1
11	AIR-TURBINE-TWO	1
12	AIR-TURBINE-THREE	1
13	AIR-TURBINE-FOUR	1
14	AIR-TURBINE-FIVE	1
15	FUEL-TURBINE-ONE	1
16	FUEL-TURBINE-TWO	1
17	FUEL-TURBINE-THREE	1
18	FUEL-TURBINE-FOUR	1
19	FUEL-TURBINE-FIVE	1
20	AIR-TURBINE-BRACKET	1
21	FUEL-INTAKE	1
22	FUEL-TURBINE-BRACKET	1
23	PRIMARY-TURBINE-CASE	1
24	BEARING-CASE	1
25	AIR-TURBINE-CASE	1
26	FUEL-INTAKE-CASE	1
27	FUEL-TURBINE-CASE	1
28	FRONT-RIM	1
29	INNER-BODY	1
30	OUTER-BODY	1



Tolerances:
.XX ± .030
.XXX ± .05
Angular ± 2°
Units are in inches.

Drawn By:
Aaryan Lath

Turbofan Jet Engine

BYPASS-ENGINE 00
7/12/25 Done on CREO

A 3D CAD model of a mechanical assembly, likely a pump or motor, mounted on a yellow rectangular base. The assembly consists of a blue main body with a green flywheel on the right side. A brown cylindrical component is attached to the top left, and a red cylindrical component is positioned in front of the base. The model is shown on a light gray grid background.

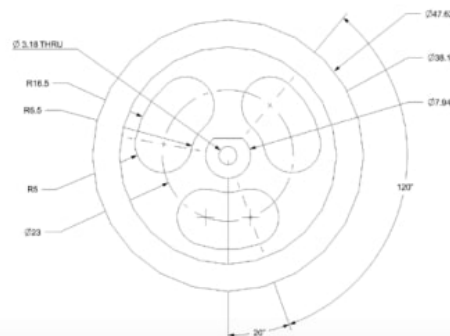
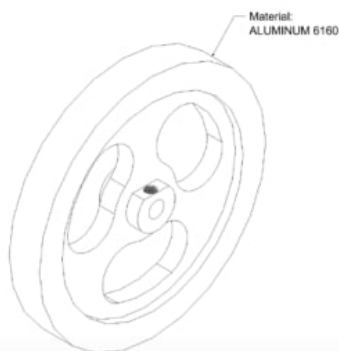
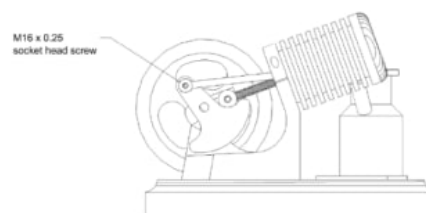
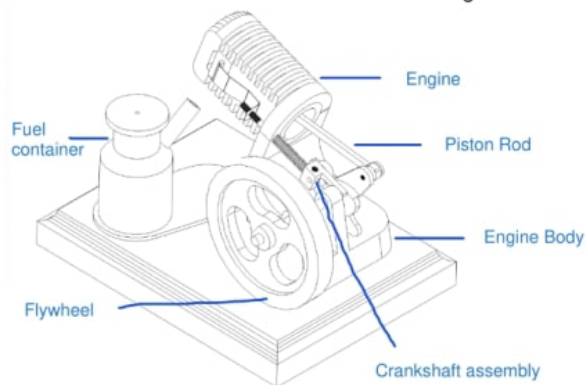
How:

- Employed **Siemens NX** for modelling and **ARAS Innovator** for **PDM** and **PLM**
- Applied **GD&T** on all components

Obtained **recognition** from Professor Travis Fuerst at **Purdue University** and worked with him for deploying **Teamcenter** the following year.

Created in NX

Single Piston Sterling Engine



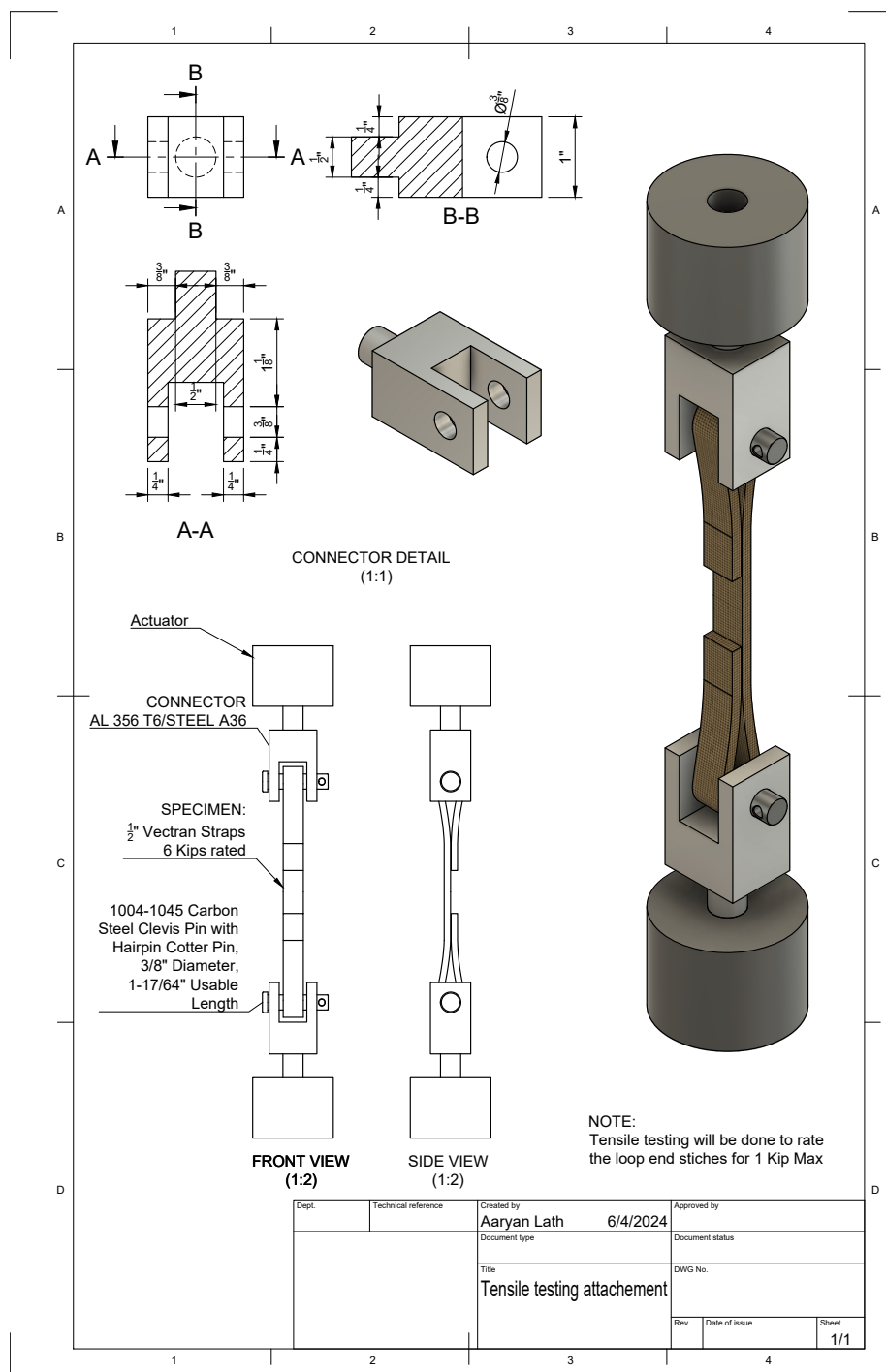
Designed and modelled an experiment for Tensile testing Vectran straps for Resilient Extraterrestrial Habitat Institute.

- Employed **Fusion 360** for modelling.

- Applied **GD&T** on all components.

- Hand-stitched the Vectran straps with different stitching patterns and **performed** a tensile test until failure.

- Conducted a **fatigue test** and developed a **rainflow counting algorithm** simulating lunar vibrations.



3D-Printing Milestones course



Designed and modelled a lightsaber as the final project for the 8 week **Fusion 360 x 3D-printing** course

How:

- Built the lightsaber using only **4** features
- 3D-printed using the **Prusa MK3 PLA 3D** printer.

